A Hybrid Deep Learning Framework for Early Lung Cancer Diagnosis and Classification Using Transfer Learning with VGG16

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Abstract. Lung cancer spreads quickly and is often detected at an advanced stage, making it one of the leading causes of cancer-related deaths globally. The nodules are often small and difficult to see, making early detection very difficult. In this work, we propose a Deep Learning (DL) method for the early diagnosis and categorization of lung cancer that uses Convolutional Neural Networks (CNNs), Transfer Learning (TL), and VGG16. Using a lung cancer imaging dataset, DL models are utilized to categorize several tumor kinds, identify malignant areas, and estimate tumor attributes including size, shape, and location. Specifically, our approach leverages pre-trained models such as VGG16 as part of the TL technique to enhance feature extraction and classification performance. In order to enhance performance, the Python-based model integrates CNNs for image classification with Recurrent Neural Networks (RNN) and Feedforward Neural Networks (FNN). The proposed method provides a more accurate diagnosis and generates tumor features that allow for more intelligent clinical decision-making. This study also highlights how important it is to create rules for the ethical use of Artificial Intelligence (AI) in healthcare.

Keywords: Deep Learning, Lung Cancer, Convolution Neural Networks, VGG16, Medical Imaging, and Transfer Learning

1 Introduction.

Globally, lung cancer is a widespread and fatal disease that kills nearly 422 people every day. Individuals above the age of 50 are prone to developing cancer, and thus the

number of patients diagnosed with lung cancer is rising on a daily basis [1]. It is believed to be among the most prevalent causes of death compared to other illnesses because it is difficult to diagnose. The main reason for the failure is that this lesion, which is a nodule, that is typically very small. The cancer cells are small at this point and grow larger as they acquire malignant traits. Managing the disease, especially at this early phase becomes essential. The chances of survival are better the sooner the illness is diagnosed. Nowadays, computer centers possess quite awesome data networks that help automate the diagnosis of identifying and separating tumorous from healthy regions [2]. Lung conditions are quickly topping the list of death causes in rich nations.

Actually, ML is a branch of AI that advanced from the research of cognitive learning principles and pattern recognition [3] [4]. Its primary objective is to create models and algorithms that can learn from and adapt to huge data sets. ML models can utilize this data to predict outcomes and make decisions by referencing past patterns and experiences [5][6]. ML algorithms may generalize and anticipate new, previously unknown information after a training step, where the model trains from annotated instances or past information. ML has become an essential tool for enhancing decision-making for a variety of companies, automating tasks, and deriving insights [7]. DL, a complex form of ML extracts and learns intricate patterns from data through multi-layered Deep Neural Networks (DNN) [8]. In many fields, DL has been found to be highly capable and efficient, sometimes even surpassing humans. AI/ML is finding increasing relevance within the healthcare sector. There has to be some regulations to test and promote AI/ML-driven software systems to predict and detect early diseases because of the pervasive usage of technologies for risk forecasting for numerous health issues [9] [10].

In this paper, we are optimizing the process of detection on the lung cancer dataset with CNN, TL and VGG16 algorithm based on ML model. DNN employ various techniques to enhance the learning and performance using the key techniques such as RNN and FNN. Employing the Python programming language for model deployment, lung cancer patients are simultaneously categorized according to their symptoms using a CNN classifier. There are several ways to diagnose various forms of cancers. But there are very few specific techniques that can actually be used to compute what populations are inside. This paper introduces the method through which not only the cancerous tumor can be diagnosed, but also to compute the size, shape, and location of them. By this, not only can the tumors be detected but one can easily know the type by counting and guessing and can also calculate the proper measures to handle them.

2 Literature Survey

It is yet to identify the actual cause of cancer and find a comprehensive cure. Early detection of cancer allows for treatment. To identify malignant parts of the lung, image processing techniques like noise mitigation, feature extraction, impacted region recognition, and perhaps contrast with data on previous clinical diagnoses of lung cancer are employed. According to S. Nageswaran et.al., in their research uses technology made achievable by ML and image processing to precisely categorize and anticipate lung cancer. Seventy different patients' 83 Computerized Tomography (CT) scans served as

the dataset for their experimental study. When preprocessing images, the geometric mean filter was applied.[12]. Through constant observation, Palani and Venkata Lakshmi [13] have provided predictive modelling of lung cancer sickness. They achieved this by using a classifier in conjunction with fuzzy cluster-linked reinforcement. The creation of precise image segmentation depends on the fuzzy clustering technique.

Nithila and Kumar built an active contouring approach with a variational level set mechanism for lung segmentation [14]. Precise segmentation of the lung parenchyma is crucial for proper analysis of lung diseases. A notable progress in CT lung image segmentation is the introduction of Selective Binary and Gaussian Filtering (SPGF) – the new Signed Pressure Force (SPF) function, which enhances the precision and effectiveness of the segmentation process in medical imaging applications. Earnest A et.al., in their work proposed few common ML techniques such as Decision Tree (DT), Random Forest (RF), Neural Network (NN), Naïve Bayes (NB), K- Nearest Neighbour (KNN), Support Vector Machine (SCVM) in Lung Cancer Prediction (LCP) [15]. Shaban [16] used an enhanced binary butterfly optimization method to develop a model for the identification of liver illness. With the greatest accuracy, precision, recall, and f-score this approach fared better than SVM, KNN, DT, and RF. Satya Prakash Maurya et al. introduced the KNN and Bernoulli NB models, which were shown to be the key methods for early LCP [17]. This research compares twelve different ML algorithms on medical records with eleven lung cancer symptoms and two significant patient habits to reliably predict a positive diagnosis.

Classification and heat map correlation were used to arrive at the conclusion. A CNN linear discrimination (LD) detection model was proposed by Mutlu et al. [18] and evaluated using the BUPA 1 and ILPD datasets. The authors also looked at how this model performed in comparison to various ML techniques, such as NB, SVM, KNN, and logistic regression (LR). The performance of this model was also contrasted with that of other ML approaches. In order to identify the best boosting strategy for predicting Chronic Liver Disease (CLD), Kalaiselvi et al. [19] looked into a number of ML approaches, including KNN and DT. Pasha et al. [20] presented the LD model for prediction and evaluated its predictive accuracy against other ML algorithms including RF, LR, and SVM. Shimazaki et al. [21] used the algorithmic approach to create and verify a DL model for the identification of lung cancer on chest radiographs. Their approach proved that it could correctly identify lung cancers on chest radiographs with a low proportion of false positives. A Computer Aided Diagnosis (CAD) mechanism based on U-Net and ResNet – 34 architecture was introduced by Silva et.al.,. They applied the method to four different cross-cohort data. Investigators used Dice Similarity Coefficient (DSC) to judge the efficacy of their approach [22] [23].

Subasi Durur Al., due to the increase in breast cancer incidence worldwide, examines the revolutionary effects of AI on breast imaging. Although adoption rates vary, AI, driven by DL techniques, offers promise in improving diagnostic procedures. Precision medicine in breast cancer research may advance as a result of its capacity to handle large datasets and process multidimensional data [24]. Authors [25] found that AI could help with breast cancer screening. However, there isn't a lot of proof that AI can be used safely in real clinical settings. In order to indicate cases for additional arbitration

review among screened women, a commercially available AI system was used as an extra reader in addition to regular double reading. BinuyaMAE, Engelhardt EG al used data from the same institution collected at different times for external validation may not be ideal. However, existing literature supports the effectiveness of temporal external validation [26], which led us to choose this approach for our study.

3 Methodology

3.1 Data Collection

We used multi-omics datasets that are publicly accessible on Kaggle to create precise models for lung cancer classification. Important clinical information such as the patient's age, gender, survival duration, and whether they had a relapse is included in one key file, known as the "Donor" file. This information is essential for forecasting. The dataset includes a variety of images that depict different lung tissue conditions, including lung adenocarcinoma, lung squamous cell carcinoma, and normal lung tissue. It was simpler to use sophisticated analysis techniques because these images had already undergone pre-processing. Based on clinical records and genetic information, we classified and diagnosed lung cancer using a combination of CNNs and DL models. Due to the scarcity of labeled data, we employed TL, which involves customizing pre-trained models to our assignment. This improved our models' performance and made it easier for them to adjust to various patient data. By using pre-trained neural networks, such as VGG16, our algorithms were able to increase accuracy and generalize more successfully across different patient profiles. We used data augmentation methods such as rotation, flipping, scaling, and color adjustment to expand the quantity and diversity of training examples. This increased the dataset's diversity and balance, which enhanced model accuracy and decreased the possibility of overfitting.

3.2 Techniques Used

CNNs are a specific type of DL model made to process images and other data with a grid-like structure. Unlike conventional fully connected neural networks, CNNs use convolutional layers with a set of learnable filters to extract spatial features from input data. These filters collect patterns at various levels of abstraction, such as textures, boundaries, and patterns, as they pass over the image. A pre-trained model created for one task can serve as the foundation for a model on a completely different but related task in TL. VGG16 was specifically used among these pre-trained models. Instead of creating a model from scratch, which necessitates a significant amount of data and processing power, TL applies knowledge from a source domain which is typically trained on enormous datasets like ImageNet to a target domain with less data. In addition to speeding up the training process, this approach usually results in improved performance, especially when dealing with small or specialized datasets.

3.3 Proposed Methodology

In this research, CNNs serve as the core DL model due to their exceptional ability to interpret high-dimensional image data. Given the complexity of lung scan images, CNNs enable automatic feature extraction directly from raw pixel inputs, making manual feature engineering unnecessary. The hierarchical structure of CNNs helps in identifying subtle patterns and abnormalities in lung tissues, crucial for early cancer detection. Before training, all images undergo pre-processing steps like scaling, normalization, and artifact removal to ensure consistency and quality. To address the issue of class imbalance in the dataset, the Synthetic Minority Over-sampling Technique (SMOTE) is applied, followed by feature selection and normalization using correlation analysis and standard scaling. The CNN architecture is then trained on this curated data to classify images as cancerous or non-cancerous. Evaluation metrics such as accuracy, precision, recall, F1-score, and AUC-ROC curve are used to assess the performance of the model, supported by k-fold cross-validation for reliability.

TL plays a pivotal role in improving the model's performance while minimizing the need for extensive training data. Since the dataset used in this project is of moderate size, TL helps leverage the learning gained from large-scale datasets like ImageNet. By adopting pre-trained models such as VGG16, ResNet50, and InceptionV3, the training process becomes faster and more efficient. These models already possess knowledge of low- to high-level features learned from millions of images, which can be fine-tuned for our specific lung cancer classification task. After pre-processing and feature engineering, the adjusted dataset is fed into these pre-trained models. The CNN layers are either frozen or partially retrained depending on the strategy, allowing the model to adapt to medical imaging nuances. TL significantly enhances prediction accuracy, enabling reliable identification of malignant cases with limited labelled data.

Among the various models evaluated, VGG16 emerged as a particularly effective architecture for lung cancer prediction due to its depth and simplicity. In this approach, VGG16 is fine-tuned using the prepared lung image dataset to harness its powerful feature extraction capabilities. The model's convolutional layers are adept at identifying complex spatial patterns in medical imaging, which are critical in detecting early-stage lung abnormalities. Leveraging VGG16 through TL ensures that the network adapts to the nuances of the domain without the need for training from scratch. The dataset is first balanced and normalized before being input into the model. The trained model is then used in a multi-step pipeline that includes prediction generation, malignant or non-malignant classification, and final result presentation. This framework contributes to a robust and intelligent system that aids clinicians in making faster, more accurate diagnoses. Fig 1 shows the proposed methodology.

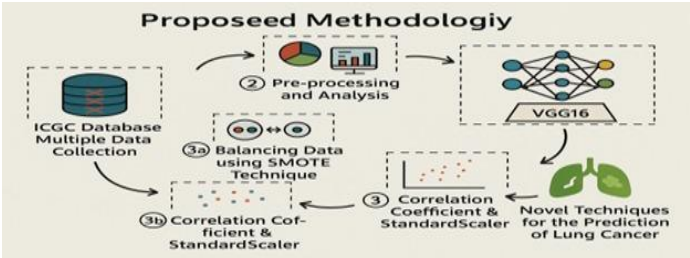

Fig. 1. Proposed Methodology

The proposed strategy for predicting lung cancer combines DL and data science techniques to increase diagnosis accuracy. First, a range of datasets are collected from the ICGC database, including clinical and genetic data. Pre-processing and analysis are used to clean, normalize, and comprehend the structure of the data. To address class imbalance, the SMOTE technique is used for synthetic oversampling. Features are then selected using correlation coefficients, and normalization is done using StandardScaler. The updated dataset is then processed via a CNN using the VGG16 architecture with TL, allowing for effective feature extraction and classification. Last but not least, the trained model uses advanced prediction techniques to accurately identify lung cancer, supporting early identification and clinical decision-making.

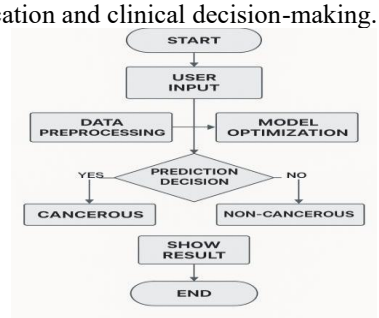

Fig. 2. Flowchart

To guarantee consistency and quality for analysis, this data passes through a data pretreatment stage. After the data has been cleaned and transformed, it moves on to the model training stage, where a predictive model is built using historical medical data. To further improve forecast accuracy and efficiency, the model is adjusted. After training, the system advances to the prediction determination phase, where it determines whether or not the input is cancerous by utilizing the model's output. If the outcome is not carcinogenic, the procedure confirms a non-cancerous conclusion; if it is, it flags it properly. When the user sees the outcome in the show result stage, the process is finally complete. This flowchart demonstrates a creative, systematic strategy to aid in early cancer detection and medical decision-making.

4 Result Analysis

In this section, three classification models—Model 1, Model 2 and Model 3 were applied to the goal of predicting subtypes of lung cancer are compared. To determine whether model, like VGG16, provides the best prediction accuracy and reliability for subtype-specific diagnosis, we compared the three models' performance in four clinically relevant domains. In addition to listing each model's advantages and disadvantages, the comparison offers details on how well-suited each model is for application in actual diagnostic scenarios. By the end of the period, the training accuracy had steadily increased from its initial level of roughly 85% to about 100%. This illustrates how well the model learns from the training dataset and keeps getting better with every epoch. The validation accuracy, however, showed a more erratic trend. It began at about 50% and increased rapidly to about 76% in the second epoch. During the training process, it fluctuated with peaks and troughs. Fig. 3 showed the highest validation accuracy (approximately 88%), indicating that this might be the best time to generalize to new data. But the subsequent drop in validation.

Fig. 3. Epoch-wise Comparison of Training and Validation Loss

Despite the fluctuations, the final validation accuracy shows improvement compared to the initial epoch, indicating that the model has still acquired significant traits from the data. Yet, the disparity in training and validation performance underlines the necessity for additional regularization procedures, such as dropout, premature stopping, or data augmentation to augment the model's generalization competence. The graph represented in Fig 4 shows the training and validation loss values across a series of epochs during the development of LCP model using TL. The training loss, (shown in blue), steadily decreases throughout the epochs, demonstrating that the model is successfully learning from the input data and curtailing its error on the training set. This consistent downward trend is typical of a well-trained model, where the learning process becomes increasingly refined over time.

Fig. 4. Training and Validation Loss for LCP Model Using TL

However, the orange line, which represents the validation loss, exhibits some variations, especially around epoch 4, where there is a noticeable increase. This anomaly points to a transient overfitting phase, in which the model performs better on training data but momentarily worse on observed validation data. Nonetheless, the notable reduction in validation loss beyond this stage implies that the model successfully modifies its parameters and enhances its capacity for generalization. This behaviour illustrates the durability of the used TL method and shows how the model might change throughout training. Two different DL strategies such as CNN model and TL method were used to assess our lung cancer detection system. Three types of lung tissue: lung adenocarcinoma (lung_aca), normal lung tissue (lung_n), and lung squamous cell carcinoma (lung_scc) were evaluated for both models using a wide range of classification metrics. A test set of 3,000 images were used to calculate the categorization results. With precision, recall, and F1-score macro and weighted average scores ranging from roughly 0.91 to 0.92, the CNN model achieved an impressive overall accuracy of 91%. With precision and recall values of 0.99 and 0.98, respectively, the model had the maximum performance in recognizing normal lung tissue (lung_n), resulting in an F1-score of 0.98. However, with F1- scores of 0.86 for lung_aca and 0.89 for lung_scc, its diagnostic efficacy for malignant cases was relatively lower. These findings point to CNN's shortcomings in differentiating across malignant subtypes, most likely because of the shared morphological characteristics present in these diseases. At 0.9 to 0.95, F1-scores were continuously high, suggesting a strong and well-rounded prediction skill. Significantly, the F1-scores for lung_scc and lung_aca both increased to 0.93 and 0.91, respectively, indicating a significant decrease in misclassifications.

This improvement can be attributed to the deeper and more abstract feature representations learned by pre-trained convolutional backbones, which are fine-tuned on the target dataset. Such architectures are particularly advantageous in medical imaging, where data scarcity and subtle inter-class differences pose significant challenges. In summary, while the CNN model demonstrates solid baseline performance, the TL approach markedly outperforms it in terms of both overall and class- specific metrics. This illustrates the benefits of using pre-trained models to progress the diagnostic precision as well as the reliability of DL-based techniques for lung cancer diagnosis.

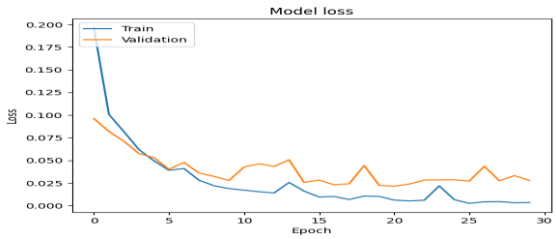

Fig. 5. Training and Validation Loss of VGG16 Model

The VGG16 model's training and validation loss curves for lung cancer prediction over 30 epochs are shown in Fig 5. Effective learning is demonstrated by the blue line, which shows the training loss, which decreases dramatically in the early epochs and progressively gets closer to zero. The orange line, which represents the validation loss, likewise decreases generally but fluctuates a little before leveling off at a low value. The consistently low loss values on both datasets suggest that the model fits the data well with little overfitting. These results validate the generalizability and reliability of the VGG16-based model in detecting lung cancer.

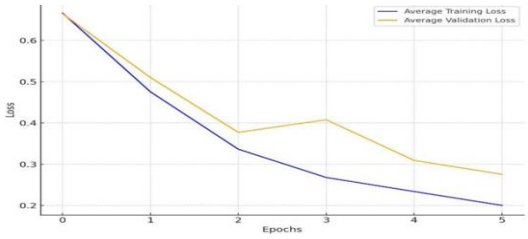

Fig. 6. Averaged Model Training and Validation Loss

Fig 6 shows the averaged training and validation loss curves from three different model training sessions. By normalizing and interpolating the loss values over epochs, the graphic provides a coherent representation of the models' learning processes. The training loss (blue), which steadily declines over time, indicates effective learning. In the meantime, the validation loss (orange), which represents possible model generalization behavior, first declines but then varies somewhat. By showcasing stability and convergence tendencies shared by all three training runs, this averaged trend provides a comparative understanding of overall training performance.

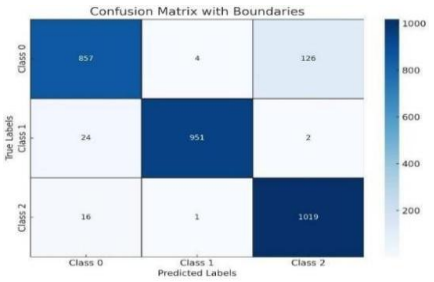

Fig. 7. CF of multi-Class Classification Model with Three Classes

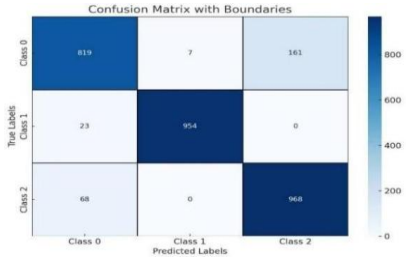

Fig. 8. Detailed CF for Three Class Classification Model

The Confusion Matrix (CF) shown in the Fig 7 represents the effectiveness of an ensemble classification model over 3 classes: Class 0, Class 1, and Class 2. Every row of the matrix correlates to the actual class labels, and each column reflects the anticipated class labels. The matrix shows that the model works well on Class 1 and Class 2, correctly predicting 954 and 968 instances respectively. There are minimal misclassifications for these classes, with Class 1 showing almost perfect accuracy. However, for Class 0, the model properly predicts 819 instances but also misclassifies 161 instances as Class 2 and 7 as Class 1, indicating some confusion between Class 0 and Class 2. Overall, this matrix provides a clear visualization of the model's benefits and weaknesses, helping to identify where improvements are needed, especially in reducing the misclassification between certain classes.

The CF presented in Fig 8 offers a complete estimation of a multi-class classification model's performance over 3 distinct classes: Class 0, Class 1, and Class 2. With the majority of cases accurately classified as 857 for Class 0, 951 for Class 1, and 1,019 for Class 2, the matrix shows a high degree of overall accuracy. When it comes to Class 2 identification, the model performs exceptionally well, with very little misclassification. Nevertheless, there is some misunderstanding between Class 0 and Class 2, as 126 instances of Class 0 are mistakenly classified as Class 2. Likewise, some Class 2 occurrences are forecasted as Class 0 (16) or Class 1 (1), and a tiny percentage of Class 1 instances are incorrectly classified as Class 0 (24) or Class 2 (2). Notwithstanding these little errors, the matrix shows that the framework is incredibly dependable and effective, making it suitable for real-world applications requiring multi-class discrimination. As a crucial diagnostic tool for comprehending classification performance and guiding future model optimization, this research highlights the CF.

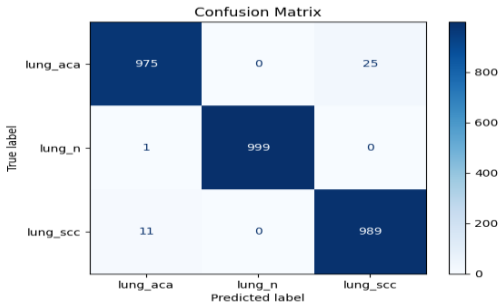

Fig. 9. Confusion Matrix of VGG16 Model for Lung Cancer Classification

The VGG16 model's classification performance across the three lung cancer classes - lung_aca (adenocarcinoma), lung_n (normal), and lung_scc (squamous cell carcinoma) is shown in Fig 9. The model correctly identified 989 lung_scc, 999 lung_n, and 975 lung_aca samples, according to the matrix. There are only a few misclassifications: 1 lung_n was projected to be lung_aca, 11 lung_scc as lung_aca, and 25 lung_aca as lung_scc. The model's efficacy and dependability in multiclass lung cancer classification are confirmed by the high values along the diagonal, which show robust predictive performance and class separation.

Table 1. Comparing Model 1, Model 2 and Model 3 Classification Metrics for Different Subtypes of Lung Cancer

Class	Metric	Model 1	Model 2	Model 3
Lung_aca	Precision	0.90	0.96	0.99
	Recall	0.83	0.87	0.97
	F1-Score	0.86	0.91	0.98
Lung_n	Precision	0.99	0.99	1.00
	Recall	0.98	0.97	1.00
	F1-Score	0.98	0.98	1.00
Lung_scc	Precision	0.86	0.89	0.98
	Recall	0.93	0.98	0.99
	F1-Score	0.89	0.93	0.98
Overall	Accuracy	0.91	0.94	0.99
	Macro.Avg	0.92	0.95	0.99
	Weighted.Avg	0.92	0.95	0.99

Based on how well each model classified the three forms of lung cancer - lung_aca (adenocarcinoma), lung_n (normal), and lung_scc (squamous cell carcinoma) - the table presents a comparison of the models. In addition to overall accuracy, macro average,

and weighted average, evaluation measures include precision, recall, and F1-score for each class. With an accuracy of 0.91, Model 1 performs admirably, although Model 2 improves with an accuracy of 0.94. With an overall accuracy of 0.99 and the greatest results across all measures, Model 3 is most likely based on the VGG16 architecture. For all three classes, it records nearly flawless precision, recall, and F1-scores, demonstrating exceptional classification skill and resilience. According to this, Model 3 is the most successful of the three in detecting lung cancer with accuracy and dependability.

5 Conclusion and Future Work

This paper proposes a trustworthy architecture for detecting and classifying lung cancer using CNN, TL and VGG16 techniques. The primary objective of the study was to assess and contrast the classification performance of conventional CNN models with TL methods, such as the use of pre-trained models such as VGG16. The basic CNN model performed well, especially when it came to accurately and reliably identifying non-cancerous lung tissue. With a weighted average F1-score of 0.92 and a total accuracy of 91%, it successfully recognized patterns in histopathology photographs. But when dealing with complicated cancer forms like adenocarcinoma, the model's efficacy somewhat declined, suggesting that it has limited generalization in situations involving complex feature distributions. By using pre-trained models to introduce TL, the accuracy of diagnostics of the system was much increased. With an improved accuracy of 94%, the TL-enhanced model (Model 2) demonstrated its ability to effectively utilize past knowledge from extensive datasets such as ImageNet. The model's improved recall and precision in malignant classes demonstrated its resilience even in the face of sparse clinical imaging data, and the performance improvement was particularly noticeable in these classes. VGG16, when used as the backbone, had the highest accuracy (99%) and consistency in measures. In extracting deep, domain-relevant characteristics, the VGG16-based method demonstrated exceptional efficacy, achieving flawless recall and precision for specific classes. This model is an effective tool for early and precise diagnosis because it reduced misclassifications and demonstrated consistent results across all lung cancer categories.

To further improve model precision, the dataset will be expanded to cover more varied lung cancer subtypes, and patient metadata will be merged. To further enhance performance and interpretability, more intricate designs like transformers and blended ensemble techniques might be investigated.

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